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Division- SY1 ENTC

Title: Implement Binary search tree with operations Create, search, and recursive traversal.

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *left;
    struct Node *right;
};

struct Node* createNode(int val){
    struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = val;
    newNode->left = NULL;
    newNode->right = NULL;
    return newNode;
}

struct Node* insert(struct Node *root, int val){
    if(root == NULL){
        return createNode(val);
    }
    if(val < root->data){
        root->left = insert(root->left, val);
    }
    else if(val > root->data){
        root->right = insert(root->right, val);
    }
    return root;
}

struct Node* search(struct Node *root, int key){
    if(root == NULL || root->data == key){
        return root;
    }
    if(key < root->data){
        return search(root->left, key);
    }
    return search(root->right, key);
}
```

```
void inorder(struct Node *root){
    if(root != NULL){
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
    }
}

void preorder(struct Node *root){
    if(root != NULL){
        printf("%d ", root->data);
        preorder(root->left);
        preorder(root->right);
    }
}

void postorder(struct Node *root){
    if(root != NULL){
        postorder(root->left);
        postorder(root->right);
        printf("%d ", root->data);
    }
}

int main(){
    struct Node *root = NULL;
    int choice, val;
    struct Node *res;

    while(1){
        printf("\n--- BST MENU ---\n");
        printf("1. Insert Node\n");
        printf("2. Search Node\n");
        printf("3. Inorder Traversal\n");
        printf("4. Preorder Traversal\n");
        printf("5. Postorder Traversal\n");
        printf("6. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        if(choice == 1){
            printf("Enter value to insert: ");
            scanf("%d", &val);
            root = insert(root, val);
        }
        else if(choice == 2){
            printf("Enter value to search: ");
            scanf("%d", &val);
            res = search(root, val);
```

```
            if(res != NULL)
                printf("Element %d found in BST\n", val);
            else
                printf("Element %d not found in BST\n", val);
        }
        else if(choice == 3){
            printf("Inorder Traversal: ");
            inorder(root);
            printf("\n");
        }
        else if(choice == 4){
            printf("Preorder Traversal: ");
            preorder(root);
            printf("\n");
        }
        else if(choice == 5){
            printf("Postorder Traversal: ");
            postorder(root);
            printf("\n");
        }
        else if(choice == 6){
            break;
        }
        else{
            printf("Invalid choice!\n");
        }
    }

    return 0;
}
```